

Interpretable Multimodal Deep Fake Detection with Cross-Attention based Fusion and Reasoning

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Abstract— The manipulated multimedia content has become indistinguishable from real content due to the advancement in deep fake generation techniques that creates problem in integrity of information and digital trust. Even though the progress is seen, the existing detection frameworks are still limited by systems and lack of interpretability which is not effective in reducing their effectiveness in real-world forensic and security applications. This paper presents an interpretable multimodal deep fake detection framework, that together analyzes variations in visual appearance, consistency of facial geometry, patterns in gaze dynamics, irregularities in boundary, speech characteristics, and the synchronization between audio and visual streams. Features extracted using pretrained models, including Vision Transformer (ViT), WavLM, SyncNet, MediaPipe Face Mesh, Gaze360, and Structured Edge Detection (SED), are fused through a cross-attention mechanism to capture complex relationships across modalities. To improve transparency, a modality-aware confidence estimation module is introduced to assess the contribution of each modality to the final prediction, while a reasoning module generates human-readable explanations of the detection outcome. The evaluation performed using the FakeAVCeleb dataset gives a strong performance of 85.2% accuracy, 0.843 F1-score, and 0.941 AUC. The evaluation results show that combining multimodal feature learning with confidence-aware reasoning improves both the detection reliability and interpretability. This work represents a step toward developing more reliable and understandable deep fake detection systems for real-world applications.

Keywords— *Deep Fake Detection, Multimodal Learning, Cross-Attention Fusion, Audio-Visual Analysis, LLM-Based Explanation Generation, Modality Confidence Estimation*

I. INTRODUCTION

Digital forensics and security systems increasingly use deep fakes due to the rapid development of generative artificial intelligence [1]. While the use of deep fake generation techniques is beneficial for media production, entertainment, and virtual content creation, its misuse in misinformation campaigns, identity fraud, or political manipulation poses significant threats to digital authenticity, information integrity, & cybersecurity. With the increasing ability of generative models that can change various aspects of human representation, such as facial appearance, geometry, gaze behavior, speech characteristics and boundary consistency, deep fakes have become more difficult to detect than in previous studies [2] [3]. The perception of these manipulations is often realistic, but they frequently result in subtle variations across visual, temporal and cross-modal

domains that can be exploited for detection [4] [5]. In spite of this, the majority of current detection methods depend mainly on unimodal analysis and only consider visual or audio information, which makes them less effective against coordinated multimodal manipulations [6] [7]. In addition, most modern deep fake detection systems operate as black-box models that only provide binary predictions without explaining why they make those decisions [8]. When people are unable to understand how the system makes its decisions, they may have less confidence in its results [9]. This can make it difficult to use such systems in important areas like digital forensics, cybersecurity, and media verification, where clear explanations and reliable decisions are essential.

A. Research Motivation

As deep fake generation methods become more sophisticated, detection systems must provide not only high accuracy but also reliable and interpretable predictions [1] [10]. In many cases, deep fake content involves simultaneous manipulations across multiple modalities, including visual, audio, and behavioral information [6] [11]. Therefore, relying on a single modality is often insufficient for capturing the complex relationships present in manipulated content. In addition, many existing frameworks focus mainly on improving detection performance while providing limited information about how the final prediction is obtained. The absence of reasoning and analysis between different modalities can improve the applicability of these systems in forensic and security-related environments [8] [12]. These challenges highlight the need for a multimodal deep fake detection framework that can effectively combine information from different modalities while also providing reliable and understandable predictions.

B. Main Contributions

This work suggests a framework for the detection of sophisticated deep fakes that incorporates information from visual, audio, and behavioral sources to solve the issue of identifying advanced forms of manipulations. By using pretrained models, the framework extracts various features, including ViT for visual appearance, WavLM for speech characteristics, SyncNet for audio and visual synchronization, MediaPipe Face Mesh for facial landmark geometry, Gaze360 for gaze behavior, and SED for boundary consistency analysis [7] [13]. As the appearance of deep fake manipulations can vary across modalities, a cross-attention based fusion mechanism is used to learn the relationship between extracted features and capture inconsistencies that may indicate

manipulated content [5] [14]. The framework has a confidence estimation module that not only determines whether viewed video is real or fake, but also assesses the influence of each modality on the final prediction. To make the results easier to understand, a post-hoc large language model (LLM) is used to generate human readable and understandable explanations from the prediction outcome and confidence scores in each modality [11]. Using data from the FakeAVCeleb dataset, the proposed framework was tested and found to have an accuracy of 85.2%, an F1-score of 0.843, and an AUC of 0.9441. The findings indicate that the use of combining information from various modalities with confidence-based reasoning can enhance the effectiveness and interpretation of deep fake detection, making the framework more appropriate for forensic and security scenarios.

II. LITERATURE REVIEW

Traditional detection methods have been unable to keep up with the rapid growth of deep fake generation techniques. Early approaches mainly focused on identifying visible distortions and inconsistencies present in manipulated images and videos [1] [2]. Despite the advancements in generative models, these irregularities have become less apparent, making detection more difficult [3] [4]. To solve this problem, researchers have explored different methods based on gaze behavior, lip synchronization, audio analysis, and visual inconsistencies. Although these methods have shown better results in specific scenarios, many focus on a single modality and do not fully capture the relationships between different sources of information. As a result, their performance may decrease when dealing with coordinated manipulations across multiple modalities.

A. Artifact Based Detection to Learned Representations

Traditional deep fake detection methods primarily utilized Convolutional Neural Networks (CNNs) and handcrafted features to detect visual artifacts like blending errors, unnatural smoothness, and compression inconsistencies [2] [10] unlike other techniques that used more complex models. Although these methods were effective in identifying earlier deep fake manipulations, they often struggled when applied to high-quality manipulated content where subtle visual artifacts were absent [2]. This flaw led to the development of representation learning methods that automatically learn discriminative features from manipulated content. A recent approach, such as Wave Guard [17], uses graph-based learning and frequency-domain analysis to enhance resilience against complex manipulations [15]. Although there are enhancements in these methods, many of them remain reliant on data-driven patterns and may not be suitable for applications that involve invisible changes. Additionally, recent studies have shown that modern generative models can suppress many traditional frequency-based artifacts, reducing the effectiveness of methods that rely heavily on spectral inconsistencies [2] [16]. Although artifact-based detection is still useful, its implementation in real-life applications may be limited by the use of only manipulation-specific traces.

B. Multimodal Deep Fake Detection Frameworks

To overcome the limitations of unimodal approaches, researchers have increasingly explored multimodal frameworks that combine visual and audio information [6] [7] [8] [11]. These methods generally achieve better performance because they utilize complementary information from multiple sources. Many existing frameworks depend on

simple feature concatenation or static fusion strategies that do not effectively model the interactions between modalities. Recent studies have introduced adaptive weighting, alignment-based learning, and audio-visual inconsistency modeling to improve multimodal learning [5] [14]. Similarly, domain-adaptive multimodal frameworks have been proposed to improve robustness across different data sets [12]. However, many approaches still rely on idealization between modalities and offer incomplete explanations of how different modalities contribute to the final prediction. As a result, cross-modal relationships remain an important area for further research.

C. Physiological and Behavioral Cue-Based Detection

Physiological and behavioral cues provide an additional layer of information for detecting manipulated content because they are often difficult to reproduce accurately in synthetic media. Research has shown that gaze behavior, eye movement patterns, facial actions, and lip synchronization can reveal inconsistencies that are difficult to identify through appearance-based analysis alone [3] [13]. Recent studies such as FauForensics [18] further demonstrate the value of incorporating facial action units into audio and visual deep fake detection. While these methods can improve detection accuracy, they are often limited by variations in pose, occlusion, lighting conditions, and image quality. Additionally, several physiological cue-based techniques evaluate these signals separately rather than jointly with visual and audio components [20]. This limits their ability to capture the complex interactions that may occur across multiple modalities in modern deep fake content.

D. Transformer Based and Spatio-Temporal Modeling

Long-range dependencies and temporal relationships that are modeled using transformers have become increasingly popular in deep fake detection. Unlike traditional CNN-based approaches, transformer architectures can learn to encode both local and global contextual information, which allows them to learn richer feature representations [7]. Recent research has revealed significant improvements in transformer-based detection through prompt tuning, multi-encoder distillation, and multimodal inconsistency learning. However, the training data and computational resources required for transformer models are typically substantial. Moreover, many current transformer-based frameworks concentrate on representation learning and feature extraction, while offering limited interpretability and inadequate modeling of cross-modal inconsistencies [5] [15] [19]. While they improve detection performance, many issues regarding multimodal reasoning, transparency and robustness remain unresolved [9].

E. Generalization and Robustness Challenges

One of the primary challenges in deep fake detection is generalization to unseen manipulations and large datasets. To improve strength [8] [10] [12] we have proposed several techniques including contrastive learning, regularization, adversarial training and even finally domain adaptation. Despite the fact that these methods can minimize overfitting and improve performance across multiple datasets, numerous models still depend on dataset-specific patterns that may not be applicable in real-life scenarios. The performance of models is often affected by unseen manipulation techniques or changing recording conditions. Moreover, the majority of current methods concentrate on improving feature representations rather than identifying the fundamental

differences between real and fake content. In high-quality applications like digital forensics and cybersecurity, calculating the accuracy of model predictions is challenging due to limited confidence estimation and explainability mechanisms [13]. To solve these limitations, there is a need for frameworks that are robust and interoperable.

F. Dataset and Evaluation Limitations

Publicly available datasets such as DFDC, FaceForensics++, Celeb-DF, and FakeAVCeleb have significantly supported deep fake detection research [11]. These datasets do not fully capture the range of manipulation techniques and real-world conditions. Therefore, models can learn from data sets only and not from the general properties of manipulated content. It is challenging to determine whether a study can truly achieve generalization accuracy since most studies use the same datasets used during training for their evaluation. Performance measurements like accuracy, precision and recall along with F1-score and AUC are commonly used evaluation metrics, but they do not always reflect the confidence or trust of model predictions. Due to these limitations, it is necessary to evaluate future deep fake detection systems using a wider range of datasets and with additional interpretability and confidence assessment mechanisms.

G. Research Gaps and Motivation

Although there are still many unresolved issues, the literature suggests that deep fake detection has made significant improvements in terms of accuracy. Complex methods based on artifacts specific detection are often ineffective, and approaches depending on single modality cannot accurately represent relationships across different information sources [6] [8]. While physiological and behavioral cues can provide useful information, they are often not analyzed together enough and may not be effective in complex situations. Similarly, transformer-based approaches for representation learning are more effective but generally less transparent and easier to interpret [9] [19]. While the existing frameworks for multimodal deep fake detection have shown better results, they often depend on simple fusion strategies where concatenating the features blindly is meaningless and offer little insight into the contribution of individual modalities [20]. According to these observations, a framework that incorporates visual, audio, and behavioral data, models cross-modal interactions effectively, while also providing interpretable predictions through confidence estimation and explanation generation.

H. Comparison of Selected Studies

Table 1 illustrates the comparison of selected studies in multimodal deepfake detection.

TABLE 1
REPRESENTATIVE COMPARISON OF SELECTED STUDIES IN MULTIMODAL DEEPPAKE DETECTION

S. No	Reference	Method	Dataset	Limitation
1.	He et al. [13]	Gaze Forensics	Visual (FF++, Celeb-DF)	Gaze-guided attention improves detection accuracy but it is limited to eye-movement features.
2.	Liu et al. [8]	Contrastive Learning	Audio-Visual (FF++)	Effective lip-speech alignment at the cost of high computational complexity.
3.	Tao et al. [16]	SAGNet	Visual (Celeb-DF)	Semantic-agnostic artifact detection with weak audio consistency modeling.
4.	Khalid et al. [11]	Multimodal Framework	Audio-Visual (FakeAVCeleb)	Large-scale multimodal benchmark lacking fine-grained synchronization labels.
5.	Haq and Malik [12]	Domain-Adaptive Representation Learning	Multimodal	Improves cross-domain robustness but provides limited interpretability.
6.	Shahzad et al. [5]	AV Lip Synchronization	Audio-Visual	Effective synchronization analysis but focuses mainly on frontal-face.
7.	Wang et al. [18]	FauForensics	Audio-Visual + Facial Actions	Utilizes facial action units but remains sensitive to variations in facial visibility and pose.

III. METHODOLOGY

This section explains the proposed interpretable multimodal deep fake detection framework, which combines audio, behavioral and visual features. The system is designed to identify important inconsistencies from different modalities that is created during deep fake generation while also providing interpretable predictions through confidence estimation and explanation generation. The proposed framework has six main stages: (i) Video Pre-processing, (ii) Multimodal Feature Extraction, (iii) Cross-Modal Attention Based Fusion, (iv) Classification, (v) Modality Aware

Confidence Estimation, and (vi) LLM Based Explanation Generation. By combining information from multiple modalities, the framework is able to capture a wider range of manipulations that may be missed when depending on a single source of information.

A. Video Pre-processing

To ensure consistent analysis of both modalities, every input video is divided into audio and visual streams before feature extraction. The preprocessing phase involves the extraction of frames and audio (Fig. 1).

1. *Frame Extraction*: Initially, the full video frames are extracted from the input video. In order to maintain a

uniform temporal coverage across all samples of video, standardized sampling methods is used by sampling 100 frames from each video. Sampled frames are cropped to obtain facial regions that removes background information and allows the model to concentrate on facial manipulations that are more suitable for detecting deep fakes [2] [13].

2. *Audio Extraction*: Each video has its own unique audio track that is standardized before feature processing. The single-channel format is used to resample audio signals up to 16 kHz, which ensures that samples at different sampling rates are equal in every instance. To minimize computational complexity, the preferred sampling rate of 16 kHz preserves the required frequency range for speech analysis [5] [6]. This output is then fed into the transformer model to extract speech features.



Fig. 1. Frame and Audio Extraction

B. Multimodal Feature Extraction

Consistencies in deep fake generation can be observed across various dimensions [1] [2], such as visual appearance, audio signals, physiological and behavioral cues. To effectively capture these variations, the proposed framework takes into account features from six supporting modalities [14], each of which helps detect a specific aspect of manipulation.

1. *Visual Appearance Feature*: Visual irregularities, including texture variations, lighting adjustments, and unnatural facial expressions can be captured using a pre-trained Vision Transformer (ViT) [9] [15]. By modeling long-range relationships across face regions, ViT distinguishes itself from traditional CNN methods and facilitates the detection of subtle indications of manipulation. To extract the 768-dimensional feature vector for every image patch, a model is constructed using cropped face frames as input [7] [15]. The transformer encoder is used to process these patch embeddings, and the [CLS] token is utilized to create a global representation for each frame. A single video-level representation is obtained by pooling temporal mean features from 100 sampled frames as shown in equation:

$$v = \frac{1}{N} \sum_{i=1}^N v_i \quad (1)$$

where v_i is the feature vector of the i -th frame and N represents the count of total sampled frames. Finally, the obtained feature is normalized to produce a robust and compact embedding for each video, that contains both local patch level details and global facial structures showing deep fake manipulations [9].

2. *Speech and Audio Feature*: Manually generated audio frequently displays subtle variations in tone, speaking rhythm, and temporal attributes that are challenging to detect [5]. These patterns are replicated using a contextual speech model that has been trained to mimic the WavLM model [5] [6]. The model divides the audio into smaller temporal portions after normalizing it to 16 kHz through normalization. A 768-dimensional feature vector is generated by WavLM transformer at each time step, representing acoustic and contextual speech information. These encompass pitch variation, speaking style, and temporal dynamics that may be modified in synthesized speech [14]. A video-level audio representation is created by pooling the time at different intervals as shown in below equation:

$$a = \frac{1}{T} \sum_{t=1}^T h_t \quad (2)$$

where h_t is the feature vector at time step t and T is the number of total audio frames. This helps in effectively identifying inconsistencies in speech that are commonly present in deep fake audio.

3. *Audio-Visual Synchronization Feature*: The most common deep fake manipulation is audio-visual inconsistency where lip movements do not match the spoken audio. Syncnet provides a shared embedding space for audio and visual inputs to learn and model the relationship between mouth movements and speech signals by producing synchronized audio and visual embedding of dimension 512. Cosine similarity is used to determine the synchronization score as shown in below equation:

$$S_{\text{sync}} = \frac{1}{N} \sum_{i=1}^N \frac{a_i \cdot v_i}{\|a_i\| \times \|v_i\|} \quad (3)$$

where a_i is the aligned audio and v_i is the visual embeddings [5] [14], respectively, and N is the number of synchronized segments. The degree of synchronization can be determined by the value of similarity, but lower values may indicate manipulation. The resulting score is important to detect inconsistencies between audio and visual mediums.

4. *Facial Landmark Geometry Feature*: Facial landmark geometry is used for studying the structural organization and movement of facial features. Face frames are processed using MediaPipe Face Mesh, which extracts a dense collection of facial landmark (x, y, z) by plotting them along three-dimensional lines [12]. Faces and movements are captured by these landmarks, which can be modified during manipulation [13]. Temporal mean pooling is done to create a 1404-dimensional feature vector and generating it for each frame to obtain video level representation as shown below:

$$l = \frac{1}{T} \sum_{t=1}^T l_t \quad (4)$$

where l_t is the landmark feature at time step t and T is the count of total frames. To obtain a consistent embedding, the aggregated feature is normalized and it effectively captures both structural and irregularities in motion of the face.

5. *Gaze Dynamic Feature*: Natural gaze patterns are challenging to replicate accurately in video manipulation, making eye gazing an effective physiological cue [3]. This information is captured by the Gaze360 model, which estimates gaze direction over video frames. Yaw is horizontal motion and pitch is vertical movement where these angles are extracted by the model over time [13], which allows for the analysis of gaze dynamics throughout the video. By means of statistical analysis, gaze behavior is summarized as:

$$g = [\mu_y, \sigma_y, \mu_p, \sigma_p] \quad (5)$$

where μ is the mean and σ is the standard deviation of yaw (y) and pitch (p) angles, respectively. The features are normalized to get a stable feature representation for each video that may expose unnatural eye movements linked to manipulated content.

6. *Boundary Consistency Feature*: Often, the manipulation of facial features is combined with the original frame, leading to subtle variations around facial boundaries [2] [16]. To evaluate these inconsistencies, edge-based features are obtained from face regions through boundary information. Edge density, edge strength, and spatial distribution around facial contours are among the extracted features. The video-level representation is constructed using statistical measures over time as shown in equation below:

$$b = [\mu_d, \sigma_d, \mu_s, \sigma_s, \mu_r, \sigma_r] \quad (6)$$

where μ and σ is the mean and standard deviation of edge density (d), edge strength (s), and spatial distribution (r), respectively [14] [15].

7. *Summary*: These six modalities together provide complementary information about visual, audio, physiological and cross-modal inconsistencies for the next fusion stage and classification stage.

C. Cross-Modal Attention Fusion:

Cross-modal attention based fusion is used to combine features from different modalities [8] [14] and capture their interrelationships. Initially, the features extracted from each modality have varying dimensions [19]. To enable effective interaction between modalities, these features are projected into a shared latent space with fixed dimensions, that allows the model to better represent and identify altered content. Let the transformed feature of each modality is represented as:

$$z_i \in \mathbb{R}^d, \quad i = 1, 2, \dots, M \quad (7)$$

where the number of modalities is M . The modalities are treated as tokens and then given to a transformer encoder for learning the relationships between modes [9] [15] and allocating importance to each accordingly. Blending the transformed features into their respective values is necessary to obtain a fused representation as illustrated below:

$$f_{\text{fused}} = \frac{1}{M} \sum_{i=1}^M z_i \quad (8)$$

where $f_{\text{fused}} \in \mathbb{R}^d$ gives the final fused feature vector, $z_i \in \mathbb{R}^d$ is the transformed feature vector of the i -th modality and M denotes the number of total modalities. Through the integration of modalities into a common feature space and attention, the model can accurately capture inter-modal relationships [8] [11].

D. Classification:

The fused multimodal representation obtained using cross-modal attention based fusion mechanism is used to classify whether the input video is fake or real [8]. To get a binary prediction, a supervised classifier is trained to generate probability along with the label that shows the manipulation likelihood. Here the prediction is computed as:

$$P(y = 1 | f) = \sigma(w^T f + b) \quad (9)$$

where f is the fused feature vector, w and b are the learnable parameters of the classifier and σ is the sigmoid function used to convert the output into a probability. To learn different types of relationships within the feature space, multiple classifiers are used, that includes Logistic Regression, Multilayer Perceptron (MLP), and XGBoost. Logistic Regression serves as a simple and interpretable baseline that models linear relationships between features and the output. while MLP have the capability of model complex nonlinear interactions among features. XGBoost further improves performance by using ensemble learning and capturing the relationships between different features effectively. The final prediction is based on a probability score, where higher values indicate a higher chance that the video is fake [16].

E. Modality Aware Confidence Estimation:

The final prediction made by the classification model doesn't directly depend on the contribution of different modalities to that decision. Improved interpretability is achieved through the usage of a confidence estimation module that assesses the contribution of each modality independently. This permits the framework to provide additional information on whether the prediction decision is influenced [7] [8] more by visual, audio, synchronization, physiological, or boundary-related characteristics. Different probability scores are generated for every modality as shown below:

$$P_m = \sigma(w_m^T e_m + b_m) \quad (10)$$

Here, e_m is the feature vector of the m -th modality, while w_m and b_m are the learnable parameters of the classifier for that specific modality, and σ is the sigmoid function. This allows the framework to estimate confidence scores for each modality separately that enhance transparency and enable users to understand the reasoning behind the final prediction more effectively than relying on a single classification result.

F. LLM Based Explanation Generation:

A post-hoc large language model that uses the final classification output as fake or real, overall prediction confidence, and modality specific confidence scores to produce human readable explanations. A structured representation of the predicted label, global confidence score of a classifier, and individual confidence calculated for each modality are used by the LLM. Afterwards, the structured data is transformed into an integrated natural language explanation that highlights the significant factors that contribute the decision, such as visual distortions, speech inconsistencies, and audio-visual synchronization mismatches.

Here, let $\hat{y}$ is the predicted label, $c = \{c_1, c_2, \dots, c_m\}$ represent the confidence scores for m modalities, and c_f is the final classification confidence. The explanation generation process can be expressed as $E = \text{LLM}(\hat{y}, c_f, c)$, where E is the generated explanation. By combining both global and each modality confidence probabilities, the LLM can able to generate explanations that show not only how strong the prediction is, but also how much each modality contributes to the final decision.

IV. SYSTEM ARCHITECTURE

The proposed system uses a multimodal architecture to extract useful features from visual, audio [8] [9], and behavioral information. As shown in Fig. 2, the framework has four main phases. First, the system separates the input video into a synchronized facial frames and audio components [10] [11]. These are then processed separately to extract features from each modality. Finally, all the features extracted are combined into a single feature representation for the final classification. Even if the audio is not clear or some video frames are having low quality, the framework can still make better predictions using information from the other modalities. The confidence estimation module also checks how much each modality contributes during prediction, which helps to improve the reliability of the final result. The explanation generation phase also makes the output easier to understand.

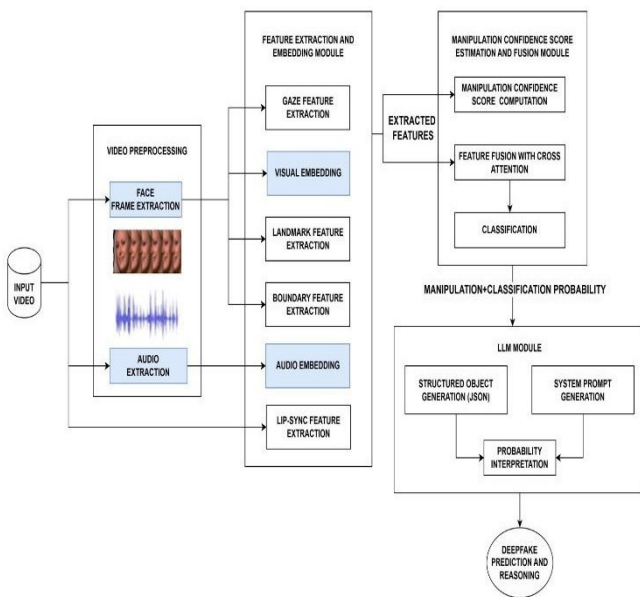


Fig. 2. Multimodal Deep Fake Detection Framework

A. Input and Pre-processing Module:

This layer processes and prepares the input video for further analysis by extracting facial frames from the video and pairing them with the corresponding audio [10] [11]. Here proper spatial and temporal alignment is maintained to keep both audio and frames are synchronized. This step helps to organize the input data into a common format, which standardizes the data and also making the later processing stages more consistent and easier to handle.

B. Multimodal Feature Extraction Module:

Feature extraction in the proposed system uses six different modalities, including visual appearance, audio characteristics, audio and video synchronization, facial landmark geometry, gaze dynamics, and boundary consistency. Each modality gives different signs of deep fake manipulation [1] [14]. For example, some features help detect visual errors, while others identify timing mismatches or unnatural facial behaviors. When all these features are used together, the system gets a more complete understanding of the input video. As the information is collected from multiple sources, the model becomes more reliable [1] [7] when handling different types of manipulation methods. This multimodal approach also reduces the limitations that can happen when depending only on a single modality and helps improve performance in difficult real world situations.

C. Fusion and Classification Module:

After the feature extraction stage, the extracted features are converted into a shared representation space and combined using a cross-attention based fusion method [8] [14]. Instead of understanding each modality separately, the model learns how all these modalities are connected with each other. This helps the system identify inconsistencies such as unnatural eye movements, unusual lip synchronization, and other inconsistent patterns [13]. Once the fusion process is completed, a classifier predicts whether the video is fake or real.

D. Interpretability Module:

To make the prediction result easier for users to understand, the framework also includes a module that tells how much each modality is involved to give the final result. Along with this, the post-hoc language model generates simple explanations for the prediction and its confidence scores that is easier to understand. Because of this, users can better understand why the system classified a video as fake or real [9].

V. EXPERIMENTAL EVALUATION

A. Dataset and Setup:

This proposed framework is tested using the FakeAVCeleb dataset, which is a large multimodal deep fake dataset that includes both visual and audio manipulations. Here four advanced techniques are used for generating voices and facial expressions, including [11] Faceswap, FSGAN, Wav2Lip which employs neural networks, and Real-Time Voice Cloning (SV2TTS), are employed in its construction. The dataset is well suitable for testing multimodal deep fake detection methods [6] [11] because these methods produce highly realistic changes in both video and audio. More than 20,000 video samples make up the data huge where diverse

collection of data available for training and evaluation. Four classifications are given based on the authenticity of the video and audio components that is (i) Real Video–Real Audio, (ii) Real Video–Fake Audio, (iii) Fake Video–Real Audio, and (iv) Fake Video–Fake Audio. This approach helps in testing different kinds of manipulations, including both unimodal and cross-modal deep fake manipulations in a more detailed way. Each video in the dataset also includes useful information like source and target identities, manipulation methods, category labels, and other file details. These annotations make the preprocessing and labeling process easier and more organized during experimentation [11]. Since the FakeAVCeleb dataset contains many types of manipulation techniques and a large number of video samples, it is useful for checking how well multimodal deep fake detection systems can handle different situations and perform on unseen dataset.

B. Performance Evaluation:

To calculate the performance of the proposed multimodal deep fake detection framework, evaluation metrics such as accuracy, precision, recall, F1-score, and Area Under the ROC Curve (AUC) are used [1]. These metrics help in understanding how effectively the model can identify real videos from fake videos [8].

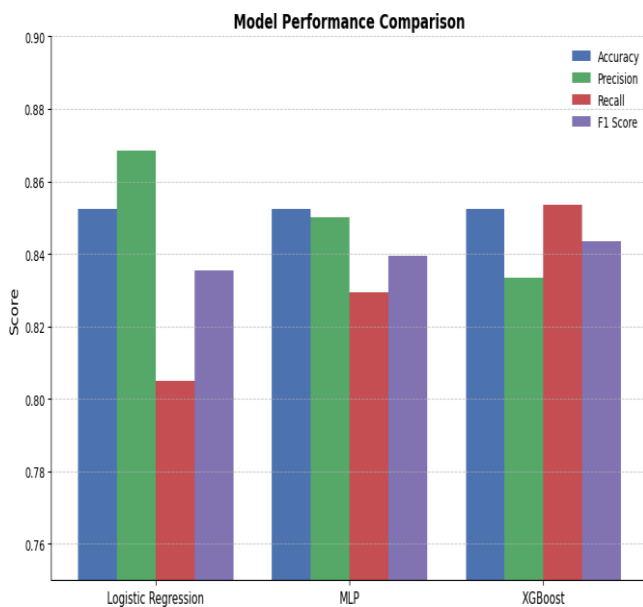


Fig. 3. Performance Comparison Between Proposed Models

As shown in Fig. 3, all classifier models such as Logistic Regression, Multilayer Perceptron (MLP), and XGBoost, achieved an accuracy of 85.2%, and also it is consistent with the experimental results. There is also some good performance from XGBoost, with the highest F1-score of 0.843 and an AUC of just 0.941, indicating that the model is working effectively in distinguishing between real and fake samples.

C. ROC Curve Analysis:

As a demonstration, the classifiers evaluation results have been plotted against ROC curves, corresponding to Logistic Regression, MLP, and XGBoost and it is shown in 4(a)–(c). The AUC values show that the classifiers are able to

distinguish fake content effectively. Overall, the results show that the proposed framework works consistently well with different machine learning and deep learning models.

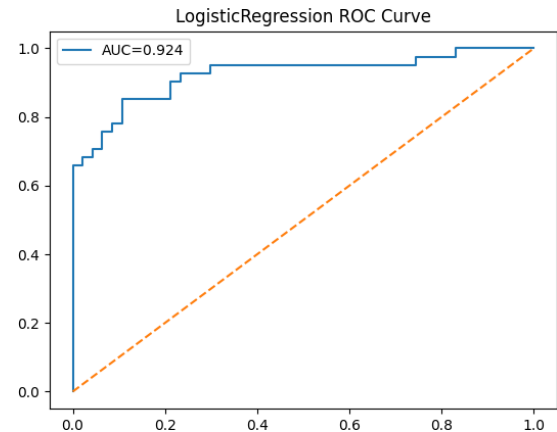


Fig. 4. ROC Curve of Logistic Regression

As seen in Fig. 4, The Logistic Regression model's AUC of 0.924 indicates that it can differentiate between real and fake samples with high precision, irrespective of the classification level. The illustration of the MLP model shown in Fig. 5 has an AUC of 0.926, which explains that it can detect nonlinear patterns in the data more effectively. This improvement shows that combining different types of features helps the model perform better. Both models are able to clearly tell the difference between real and fake videos, with the MLP performing slightly better than the basic linear model. Overall, the results show that using multiple features together improves performance, and the MLP gives a small boost compared to the baseline.

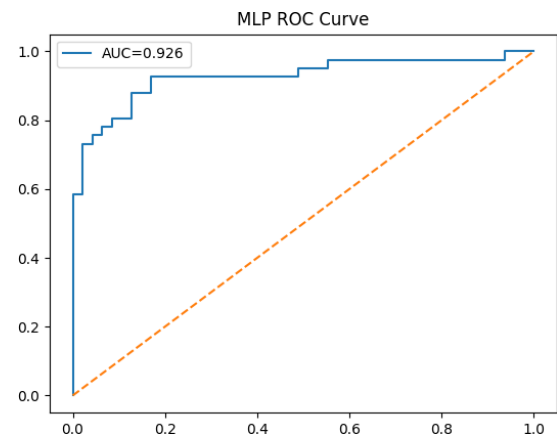


Fig. 5. ROC Curve of MLP

The ROC curve for XGBoost is shown in Fig. 6 and the maximum AUC is 0.941. At a lower false positive rate than other models, the true negative rate of the curve is still higher when compared to other ROC curves. The result shows that XGBoost is working better in detecting fake videos more accurately particularly in challenging manipulations, which helps in reducing false positive predictions. This shows that the model is still able to give better results even when the fake videos contain difficult visual and audio manipulations. The

ROC curve also suggests that XGBoost can identify real and fake samples more clearly at different threshold values. Since the model gives consistent performance throughout the experimentations, XGBoost can be considered a reliable classifier for the proposed deep fake detection system.

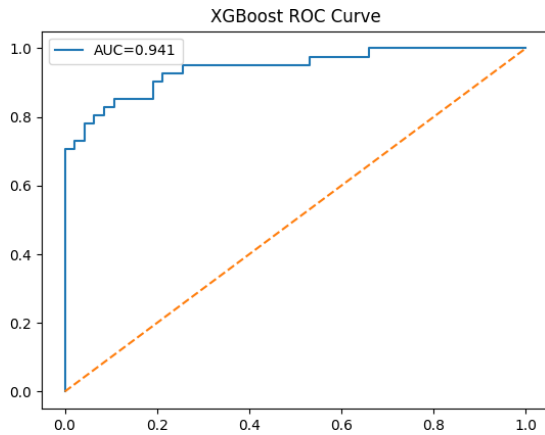


Fig. 6. ROC Curve of XGBoost

D. Confusion Matrix Analysis:

The confusion matrices for each of the classifiers which are Logistic Regression, MLP and XGBoost, that have been evaluated is illustrated in fig 7-9.

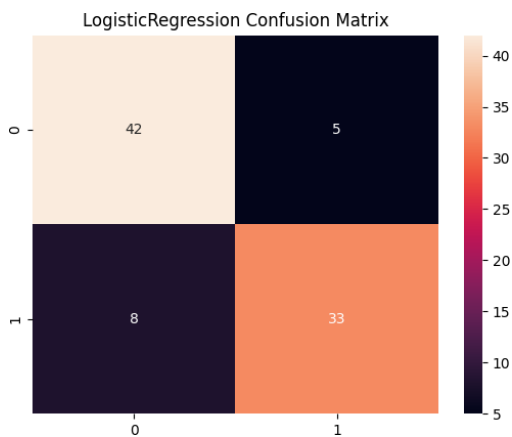


Fig. 7. Confusion Matrix of Logistic Regression

As shown in Fig. 7, Logistic Regression contain 32 real positive integers and so it is clear that model is capable of accurately identifying a large proportion of manipulated videos. Additionally, the confusion matrix of Logistic Regression also indicates that it is capable of capturing significant deep fake characteristics.

A confusion matrix for the MLP model is shown in Fig. 8. It identifies more true positives, which improves detection performance. The model learns complex patterns in the data and can detect manipulated samples more accurately where the system distinguishes real and manipulated samples with greater reliability. As a result, the MLP model achieves a more balanced classification performance and reduces the number of incorrect predictions.

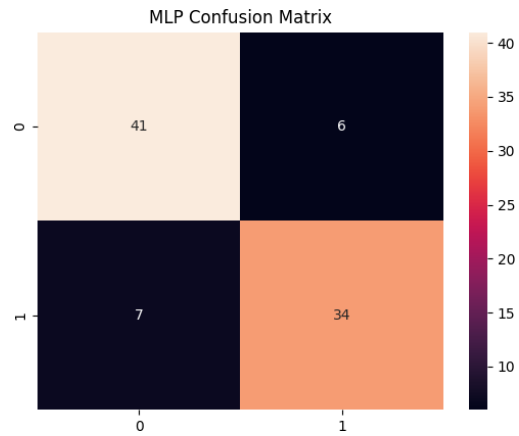


Fig. 8. Confusion Matrix of MLP

Fig. 9 shows that XGBoost achieves the highest number of true positives (35). This indicates that it detects deep fakes more effectively than the other methods, likely due to its ability to capture interactions among multimodal features.

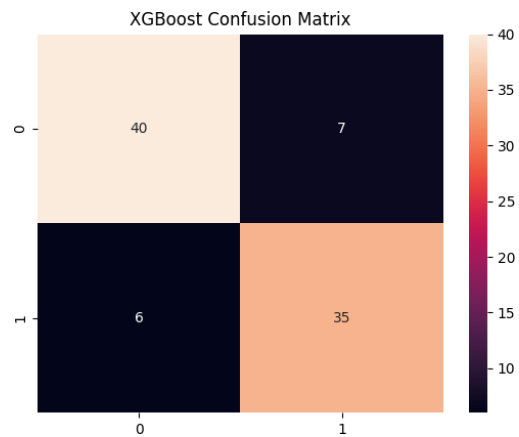


Fig. 9. Confusion Matrix of XGBoost

XGBoost performs better than the other models because it detects more manipulated samples. Based on this, it can be concluded that XGBoost provides the best overall performance in this study.

E. Result Analysis:

Experimental results clearly indicate that the proposed framework effectively uses the multimodal fusion for effective deep fake detection. Even achieving highest accuracy of 85.2%, it is still a challenge to detect deep fakes that are highly realistic, and the AUC value of 0.941 indicating a strong ability to differentiate between real and fake content across different decision thresholds. The size of the dataset can also affect the performance of the model [10] [12]. Because of this, the model may sometimes struggle to handle more complex manipulations that are not seen often during training. Even with these limitations, the results still show that combining multimodal feature extraction with interpretability methods can give reliable and scalable deep fake detection performance [1] [7].

When the proposed work is compared with existing methods, a few clear differences can be noticed. Some earlier studies do not include AUC [19] as an evaluation metric, so it becomes harder to understand how effectively those models used in the studies can distinguish between real and fake videos at different threshold values. Also, some frameworks show lower accuracy, especially they depend only on single modality features [3] [4]. In addition, many existing methods do not use multimodal feature fusion, limiting their ability to capture complex deep fake patterns. In contrast, the proposed framework uses multimodal features, showing better detection performance and improved generalization.

F. Ablation Study:

An ablation study helps to understand how much each modality has contributed. This help to identify which modality is playing a stronger role in detecting a video as fake or real. To do this, individual models are trained using each modality separately, and their performance is compared with the full multimodal framework where it uses a fused embedding from all six modalities that is considered from a video.

TABLE 2

ABLATION STUDY SHOWING PERFORMANCE OF INDIVIDUAL MODALITIES AND THE FULL MULTIMODAL FRAMEWORK

Modality	Accuracy	F1	AUC
Visual	0.5455	0.5122	0.6015
Gaze	0.5682	0.5366	0.5739
Boundary	0.5341	0.5060	0.5413
Landmark	0.5114	0.4691	0.5340
Audio	0.7045	0.6905	0.8012
Sync	0.6932	0.6197	0.7343
Fused Vector	0.852	0.843	0.941

1. *Individual Modality Performance:* In general, models using a single modality perform at less than the full system as seen in Table 2. Spatial and geometric features are not enough for accurate detection of deep fakes since the accuracy and AUC values are relatively low because not most of the videos having these features manipulated where the manipulations repeatedly arise in other modalities. The most effective way to detect manipulated content is through audio, followed by audio and visual synchronization, which shows the importance of temporal consistency and analyzing the speech related information over time.
2. *Effect of Multimodal fusion:* Multimodal fusion has a strong impact, as its performance is better than all the unimodal setups. It achieves better performance, which shows that combining features from different sources helps capture minor and varied manipulation patterns. Also the lower performance of individual modalities can also be explained by the nature of deep fake data. In many cases, not all parts of a video are manipulated at the same time. For example, some videos may have fake

audio while the visual content remains real. In such situations, if a model depending only on the visual modality, it may correctly classify the video as real, even though the audio is fake. This leads to a drop in overall performance. A similar problem is seen across all six modalities, where each modality alone may fail to detect some particular types of manipulation. It highlights the importance of combining multiple modalities to achieve more accurate and reliable detection.

3. *Comparative Analysis:* The performance of unimodal systems is lower when compared to multimodal systems, showing that using only one modality may not be enough for accurate deep fake detection. When different modalities are combined, audio and synchronization features provide strong clues of manipulation, while visual appearance, gaze behavior, and boundary consistency gives extra information that helps to improve the overall performance of the system. At the same time, single modality analysis is still useful because it helps in understanding how much each modality affects the final prediction. This makes the framework not only effective in detecting deep fakes but also easier to understand and interpret.
4. *Summary:* The ablation study results show that every modality gives useful information to the final prediction. From the final model analysis, it can be seen that the cross-modal fusion method works well in combining features from different sources. The results also show that deep fake detection becomes more accurate when visual, temporal, and behavioral features are analyzed together instead of using only a single type of feature.

VI. CONCLUSION AND FUTURE WORK

A multimodal framework for deep fake detection that incorporates visual, audio, and behavioral information is presented in this paper. This framework can be used to identify manipulation introduced during the process of creating deep fake media. Feature extraction from six supporting modalities is achieved by using a cross-attention-based fusion mechanism, which allows the proposed framework to capture subtle inconsistencies that may not be detected by other sources of information alone. By analyzing the FakeAVCeleb dataset, it was found that the proposed approach is highly effective due to its 85.2% accuracy, F1-score of 0.843, and AUC of 0.9441. Overall, XGBoost performed the best among classifiers in terms of learning complex relationships within the fused multimodal feature space. Modality-aware confidence estimation and LLM-based explanation generation are utilized by the framework to enhance interpretability, as well as detection performance. Even with these encouraging findings, certain restrictions remain. The effectiveness of the framework may be influenced by both the diversity and size of training data, which can impact its generalization to methods of manipulation that are not visible to humans. Additionally, multiple feature extraction models require a higher computational load, which makes it more difficult to deploy in real-time. Upcoming work will focus on improving generalization from larger, diverse datasets, testing the framework against multiple benchmark dataset and developing faster computationally efficient end-to-end

architectures. By utilizing attention-based explanation techniques and other behavioral cues, interpretability may be enhanced by comprehending sophisticated deep fake manipulations more effectively. These improvements can contribute to the development of deep fake detection systems that are more robust, transparent, and suitable for practical forensic and security applications.

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